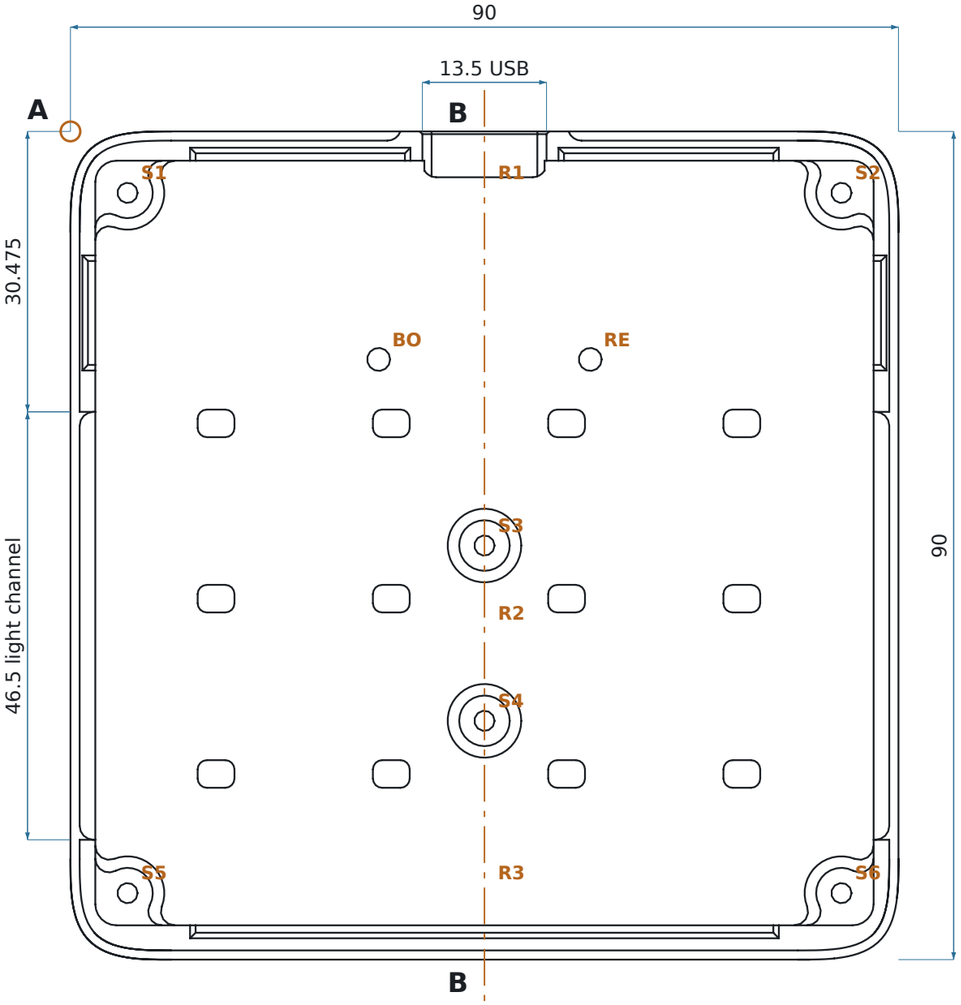
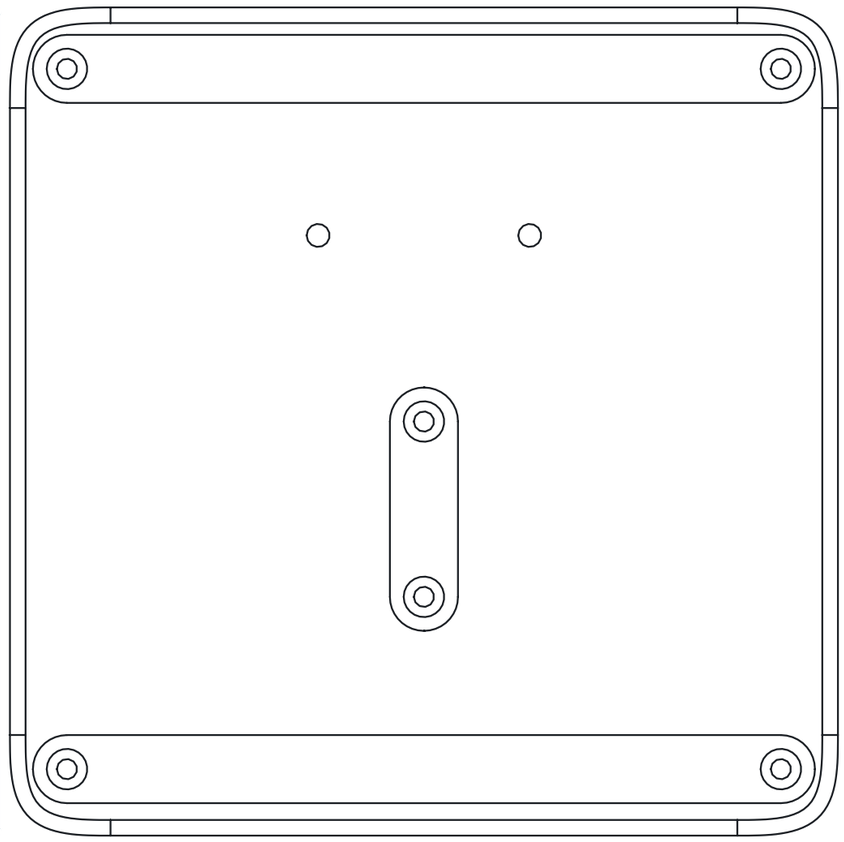
FROM ABOVE



FROM BELOW (mirrored left to right)

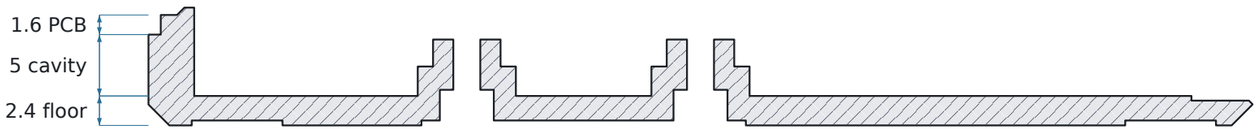


macropad case: base, CNC variant

Holes and recesses, from datum A

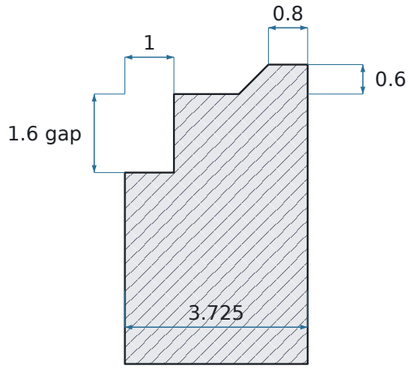
tag	X	Y
⌀2.2 through; counterbore ⌀4.4 to 2.9 from below		
S1	6.2	6.675
S2	83.8	6.675
S3	45	45
S4	45	64.05
S5	6.2	82.775
S6	83.8	82.775
⌀2.5 through (BOOT)		
BO	33.5	24.775
⌀2.5 through (RESET)		
RE	56.5	24.775
slot 85 x 7.4, full-radius ends, 0.4 deep from below		
R1	45	6.675
slot 7.4 x 26.45, full-radius ends, 0.4 deep from below		
R2	45	54.525
slot 85 x 7.4, full-radius ends, 0.4 deep from below		
R3	45	82.775

SECTION B-B



Scale 1.5:1 plans, 2:1 section, 8:1 detail, on A3. Dimensions in mm, computed from the model.
Datum A: back-left corner of the box round the outline, seen from above. X to the right, Y toward the front.
3-axis, two setups: inside up, then underside up.
Every pocket corner is R1.5 or larger: a 3 mm end mill reaches them all.
45 degree chamfers: 1.7 on the underside rim, the flank and ends of the lap ridges, 0.3 on the USB lead-in.
The outer skin's top edge at the gap stays sharp, and so does the rim inside the gap: the top seats on it.
The rim steps in 1 mm over the top 1.6 mm, all round: that is the gap. Cut it with the side of the end mill in setup 1.
Screw holes go straight through: M2 x 8 socket head from below.
Outline corners: one quintic Bezier each, control points 10.94, 6.56 and 2.19 mm from the box corner on both sides. Take the curve from the STEP file.
General tolerance: ISO 2768-m is a proposal. Nothing has been made or measured.

GAP AND LAP JOINT, front wall, 8:1



MATERIAL AND FINISH (a proposal; nothing has been made)
6061-T6 aluminium.
Fine bead blast, 120 to 150 grit glass bead, then Type II sulphuric anodise, 10 to 15 um (a guess), dyed clear (Silver) or grey (Space grey).
Anodise grows each face by about 0.01 to 0.02 mm (a guess). Every fit on this sheet leaves 0.1 mm or more per side, so no fit is cut to allow for it.
Break unmarked edges 0.05. The 0.3 chamfers are cosmetic: measure them after blasting.
WORKHOLDING
Setup 1: cut the light channels last, after the cavity is roughed: 3 mm 3-flute end mill, 0.2 mm radial steps, full depth in two passes. Each channel leaves a free 1.0 mm skin.
Setup 2 (underside up): sit the base on a plug fixture that fills the cavity, 0.1 mm smaller all round and 4 mm tall, so it backs up both skins. Hold it with an M2 screw through each of the six holes into the fixture, or in soft jaws cut to the outline below the gap. Never in a plain vise.